

Project B3: Centralized Curriculum Management System

Project Statement

Build a comprehensive curriculum management platform that allows educational institutions to define, organize, and maintain their complete academic structure digitally. The system must support multi-level hierarchy: programs → courses → subject modules → topics → learning objectives. Each topic should support attachment of digital study materials, assignments, and scheduling timelines. AI should assist in auto-sequencing topics based on learning dependencies, identifying gaps in curriculum coverage, and suggesting improvements based on standard academic frameworks.

Expected Output

Frontend:

- Drag-and-drop curriculum builder interface to define and reorder programs, courses, modules, and topics
- Rich-text and file-upload support for study materials per topic
- Calendar/timeline view for academic scheduling
- AI-powered curriculum gap analysis report view
- Search and filter functionality across the entire curriculum repository
- Role-based views: admin builds curriculum, teachers see assigned subjects, students see enrolled courses

Backend:

- CRUD APIs for the full curriculum hierarchy
- File management system for study materials
- AI-assisted topic sequencing engine based on prerequisite logic
- Curriculum comparison tool to benchmark against standard syllabi
- Academic timeline management APIs with conflict detection (e.g., overlapping sessions)
- Audit logs for any curriculum change

Deployment:

- Multi-tenant support (multiple institutions using same platform)
- Version control for curriculum changes (rollback capability)

- Secure, role-based access control across all API endpoints

Note to interns: These projects are intentionally scoped at production-system complexity. You are expected to make all architectural, technical, and design decisions independently — including how to handle scale, latency, data privacy, model selection, and infrastructure. Ability to research, justify, and defend your choices is part of the evaluation.